

AI-Enhanced Cybersecurity Threat Detector

Home Lab Project Documentation · v1.0

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1. EXECUTIVE SUMMARY

The AI-Enhanced Cybersecurity Threat Detector is a fully self-contained home lab project that combines machine learning, open-source security tooling, Microsoft's enterprise security stack, and AI-assisted reporting to simulate a real-world Security Operations Center (SOC) pipeline on a single machine. The project uses Docker containers to generate, capture, and analyze network traffic, then feeds detected anomalies through both Splunk SIEM and Microsoft Sentinel, automates response via XSOAR, enriches endpoint telemetry through Microsoft Defender for Endpoint, and produces actionable CTI incident reports via the Claude API.

This project was designed to serve three goals simultaneously: build hands-on technical depth across the full threat detection lifecycle, produce a portfolio artifact that demonstrates enterprise-grade thinking to hiring managers, and close specific skill gaps in Microsoft Defender and Sentinel that are increasingly required in modern SOC roles.

Version 2.0 of this documentation adds the Microsoft Azure, Sentinel, and Defender stack as a cloud-connected layer alongside the existing local Docker environment, enabling dual-SIEM operation and endpoint telemetry collection.

2. WHY WE ARE BUILDING THIS

2.1 Career Relevance

The cybersecurity job market increasingly demands candidates who can demonstrate hands-on experience with the full SOC stack. This project spans threat simulation, detection, dual-SIEM analysis (Splunk + Sentinel), SOAR automation, endpoint telemetry via Defender, and AI-assisted reporting — making it directly relevant to roles in CTI, detection engineering, and security operations.

Critically, Microsoft Defender and Sentinel are explicitly listed requirements in target roles such as the Primerica Information Security Analyst position. This project provides concrete, demonstrable hands-on experience with both platforms.

2.2 Skill Development

- **AI + Security:** Applying transformer models to network anomaly detection is an emerging discipline. Building this project positions you at the intersection of two high-demand fields.
- **Dual-SIEM Proficiency:** Operating Splunk and Microsoft Sentinel simultaneously mirrors real enterprise environments undergoing SIEM migrations and provides fluency in both platforms.
- **Microsoft Security Stack:** Hands-on Defender for Endpoint deployment and Sentinel workspace configuration directly addresses the Microsoft-specific skill gaps identified in resume reviews.
- **Enterprise SOAR:** XSOAR playbook automation integrated with both Splunk and Sentinel mirrors the tooling used in professional SOC environments including Leidos.
- **CTI Report Writing:** Integrating Claude as an automated report generator reinforces structured intelligence writing skills developed through the AttackIQ course and Leidos internship.
- **Cloud Security:** Azure deployment experience adds cloud security context that is increasingly required across all security roles.

2.3 Portfolio Impact

A GitHub repository containing a working, documented home lab with a docker-compose.yml that spins up the entire local environment in one command — alongside Azure-connected Sentinel and Defender integration — signals both technical depth and enterprise awareness. It is a concrete artifact interviewers can examine, clone, and run.

2.4 Future Expandability

The architecture is designed to scale. The Microsoft stack integration is already cloud-connected, meaning the project naturally evolves toward a hybrid on-premises/cloud SOC architecture. When hardware allows, the local simulation layer migrates to Proxmox VMs with zero changes to the detection and response stack.

3. PROJECT OVERVIEW

3.1 Objective

Develop a system that uses transformer-based machine learning models to analyze network traffic captured from a simulated Docker home lab, detect anomalies, and automatically trigger a response pipeline through both Splunk SIEM and Microsoft Sentinel — with endpoint telemetry from Microsoft Defender for Endpoint enriching the dataset — culminating in AI-generated CTI incident reports via the Claude API and automated case management in XSOAR.

3.2 Technical Problem Solved

Traditional signature-based detection misses novel attack patterns. This project implements behavioral anomaly detection using sequence-modeling transformers (BERT-style) trained on network traffic features, enabling detection of previously unseen threat patterns. Adding Defender for Endpoint enriches detection with host-based telemetry that network traffic alone cannot provide.

3.3 Scope

- Network traffic simulation between attacker and victim Docker containers
- Packet capture and preprocessing via Wireshark and feature engineering
- Transformer model training and inference for anomaly scoring
- Dual-SIEM alert ingestion: Splunk Free (local) and Microsoft Sentinel (cloud)
- Microsoft Defender for Endpoint deployed on victim container for endpoint telemetry
- Automated playbook execution in XSOAR Community Edition with Sentinel integration
- IOC enrichment via VirusTotal and AbuseIPDB integration
- AI-generated CTI incident reports via the Claude API
- React-based frontend dashboard for real-time monitoring

3.4 Out of Scope (Phase 1)

- Production-grade hardening of the lab environment
- Multi-machine Proxmox deployment (reserved for future Phase)
- Mobile application or external API access
- Full Azure infrastructure deployment (Sentinel and Defender are the primary Azure touchpoints)

4. ARCHITECTURE

4.1 Architecture Diagram

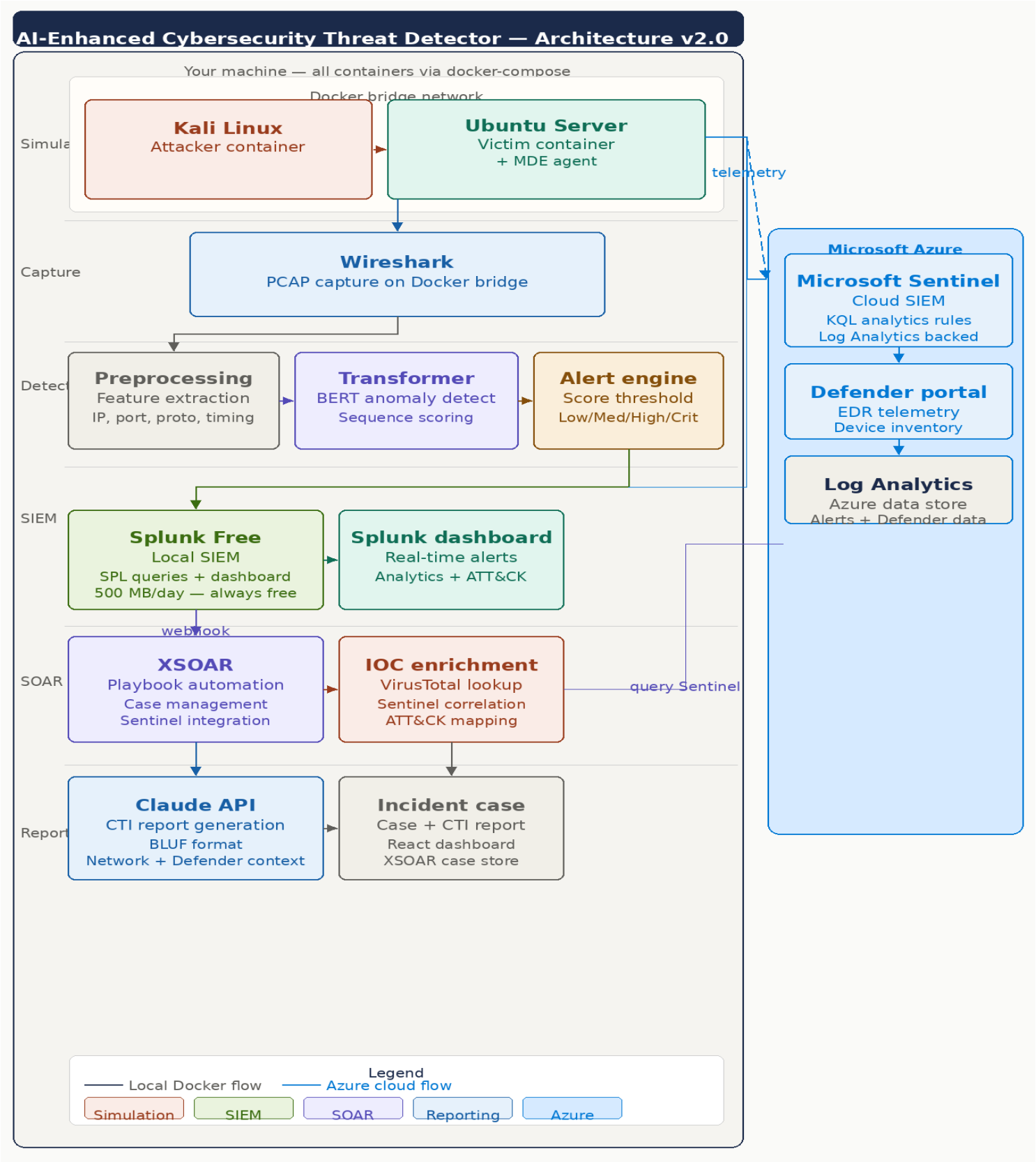


Figure 1 — Full single-machine + Azure hybrid architecture (v1.0). Left: Docker containers on your machine. Right: Microsoft Azure cloud layer (Sentinel, Defender, Log Analytics). Dashed lines indicate Azure cloud data flows.

4.2 Deployment Model

The project operates as a hybrid local/cloud architecture. All simulation, detection, and orchestration services run as Docker containers on a single host machine, orchestrated by docker-compose. Microsoft Azure hosts the Sentinel workspace and Defender for Endpoint integration, connected to the local environment via HTTP Event Collector and the Microsoft Monitoring Agent.

4.3 Pipeline Layers

Layer	Components	Purpose
Simulation	Kali Linux container, Ubuntu Server container	Generate realistic attacker/victim network traffic
Capture	Wireshark (host), tshark, PCAP export scripts	Capture and export raw network packets
Endpoint	Microsoft Defender for Endpoint on victim container	Host-based telemetry: processes, files, network connections
Detection	Preprocessing service, Transformer model API, Alert engine	Extract features, score anomalies, fire alerts
SIEM — Local	Splunk Free, Splunk dashboard	Ingest alerts, correlate events, visualize threats locally
SIEM — Cloud	Microsoft Sentinel, Azure Log Analytics	Cloud-native SIEM with Defender data connector
SOAR	XSOAR Community Edition, IOC enrichment	Automate response playbooks, enrich indicators
Reporting	Claude API, Incident case store	Generate CTI reports, close incident cases
Frontend	React dashboard	Real-time visualization of alerts and incidents

4.4 Microsoft Stack Integration Detail

The Microsoft components interact with the rest of the pipeline at three specific points:

- **Defender for Endpoint on victim container:** MDE agent installed on the Ubuntu Server victim container generates endpoint Detection and Response (EDR) telemetry — process creation events, file modifications, network connection logs, and alert signals. This data streams directly into the Azure Log Analytics workspace that backs Sentinel.
- **Sentinel as parallel SIEM:** The same alert events forwarded to Splunk via HEC are simultaneously forwarded to Sentinel via the Azure Monitor HTTP Data Collector API. Sentinel also receives Defender telemetry via its native Defender connector. This creates a dual-view of every incident across both SIEM platforms.
- **XSOAR Sentinel integration:** XSOAR's Microsoft Sentinel content pack enables playbooks to query Sentinel incidents, pull Defender alerts, and update Sentinel incident status — extending XSOAR orchestration across both SIEM platforms.

4.5 Data Flow

1. Kali Linux container executes simulated attack techniques against the Ubuntu Server container across the Docker bridge network.
2. Wireshark captures network traffic on the Docker bridge interface and exports rolling PCAP files.
3. Microsoft Defender for Endpoint on the Ubuntu Server container simultaneously captures host-based telemetry (process events, file writes, network connections) and streams it to Azure Sentinel.
4. The preprocessing service parses PCAP files, extracts features (IPs, ports, protocols, packet sizes, timing, TCP flags), and normalizes the data.
5. The transformer model API scores each traffic sequence. Sequences exceeding the anomaly threshold trigger alert events.
6. Alert events are forwarded in parallel to: (a) Splunk via HTTP Event Collector and (b) Microsoft Sentinel via Azure Monitor HTTP Data Collector API.
7. SPL queries in Splunk and KQL queries in Sentinel both correlate events and populate their respective threat dashboards.
8. A Splunk webhook triggers an XSOAR playbook on High/Critical alerts. The playbook queries Sentinel for correlated Defender telemetry to enrich the incident.
9. The XSOAR IOC enrichment playbook queries both VirusTotal (general reputation) and AbuseIPDB (IP-specific abuse confidence score, report count, ISP, and usage type) for each extracted indicator.
10. A final playbook step calls the Claude API with the full enriched context — network anomaly score, Defender telemetry, Sentinel correlated events, VirusTotal IOC data, and AbuseIPDB abuse scores — and receives a structured CTI incident report.
11. The completed incident report and case metadata are stored in XSOAR and displayed on the React frontend dashboard.

5. TOOLS & SOFTWARE

5.1 Simulation Layer

Tool	Version / Edition	Purpose	Cost
Docker Desktop	Latest stable	Container orchestration host	Free
docker-compose	v2+	Multi-container orchestration	Free
Kali Linux	Official Docker image	Attacker container OS	Free
Ubuntu Server	22.04 LTS Docker image	Victim container OS	Free
Scapy	Python library	Custom packet generation scripts	Free

5.2 Capture Layer

Tool	Version / Edition	Purpose	Cost
Wireshark	Latest stable	Packet capture and PCAP export on host	Free
tshark	Bundled with Wireshark	CLI-based capture for automation scripts	Free
tcpdump	Built into Kali image	Lightweight in-container capture	Free

5.3 Endpoint Layer - Microsoft Defender for Endpoint

Tool	Version / Edition	Purpose	Cost
Microsoft Defender for Endpoint	P1 / Trial	EDR telemetry from victim container	30-day free trial
MDE Linux Agent	Latest	Agent installed on Ubuntu Server container	Included with MDE
Azure Log Analytics Workspace	Standard tier	Receives and stores Defender telemetry	First 31 days free via Sentinel trial
Microsoft Defender Portal	cloud.microsoft.com	Alert management and device inventory	Included with MDE

Note: Defender for Endpoint requires an Azure subscription. Use the 30-day free trial. After trial, MDE P1 is approximately \$3/device/month.

5.4 Detection Layer

Tool	Version / Edition	Purpose	Cost
Python	3.11+	Core language for ML pipeline	Free
PyTorch	2.x	Transformer model training and inference	Free
Hugging Face Transformers	Latest	Pre-trained BERT model base	Free
FastAPI	Latest	Model serving API container	Free
scikit-learn	Latest	Preprocessing and evaluation metrics	Free
pandas / numpy	Latest	Data manipulation and feature engineering	Free
UNSW-NB15 Dataset	Public	Primary training data for anomaly detection	Free
CICIDS2017 Dataset	Public	Supplemental training data	Free

5.5 SIEM Layer - Splunk (Local)

Tool	Version / Edition	Purpose	Cost
Splunk Free	9.x	Log ingestion, SPL queries, local dashboards	Free — 500 MB/day
Splunk HEC	Built-in	HTTP Event Collector for alert ingestion	Free
Splunk Docker image	Official	Containerized Splunk deployment	Free

5.6 SIEM Layer - Microsoft Sentinel (Cloud)

Tool	Version / Edition	Purpose	Cost
Microsoft Azure Account	Free tier	Cloud subscription for all Azure services	\$200 credit / 30 days free
Microsoft Sentinel	Standard	Cloud-native SIEM and SOAR platform	31-day free trial, then ~\$2.46/GB ingested
Azure Log Analytics	Workspace	Data store backing Sentinel	Included with Sentinel trial
Sentinel Defender Connector	Built-in	Native connector for MDE telemetry	Included with Sentinel
Azure Monitor HTTP API	Built-in	Receives alert forwarding from local detection layer	Included with Log Analytics
KQL (Kusto Query Language)	N/A	Query language for Sentinel analytics rules	Free

Note: Activate the Sentinel free trial only when you reach Phase 3 to maximize the 31-day window during active development.

5.7 SOAR Layer

Tool	Version / Edition	Purpose	Cost
XSOAR Community Edition	8.x	Playbook automation and case management	Free
XSOAR Sentinel Content Pack	Built-in	Enables XSOAR playbooks to query Sentinel	Free
XSOAR Splunk Integration	Built-in	Webhook trigger and data pull from Splunk	Free
VirusTotal API	v3 free tier	General IOC enrichment — files, hashes, IPs, URLs	Free — 500 req/day
AbuseIPDB API	v2 free tier	IP-specific abuse reputation scoring	Free — 1,000 req/day
MITRE ATT&CK Navigator	Web app	Technique mapping for playbook tagging	Free

5.8 Reporting & Frontend Layer

Tool	Version / Edition	Purpose	Cost
Claude API (Anthropic)	claude-sonnet-4-6	AI-generated CTI incident reports	Usage-based ~\$5-20/month
React.js	18+	Frontend threat dashboard	Free
Recharts / Chart.js	Latest	Data visualization components	Free
Node.js / Express	LTS	Backend API for dashboard	Free

5.9 Developer Tools

Tool	Version / Edition	Purpose	Cost
Claude Code	Latest	AI-assisted coding from terminal	Free with Anthropic API key
Git	Latest	Version control	Free
GitHub	Free tier	Repository hosting and portfolio	Free
VS Code (optional)	Latest	Editor alongside Claude Code	Free

5.10 Total Cost Summary

Period	Estimated Cost	Notes
Setup (Month 1)	\$0 — \$10	Azure \$200 credit covers all Azure costs. Claude API ~\$5-10 during initial build. VirusTotal and AbuseIPDB free tier.
Active development (Month 2)	\$5 — \$25	Sentinel trial active. MDE trial active. Claude API usage increases.
Post-trial ongoing	\$10 — \$35/month	Sentinel ~\$5-15/month at home lab volumes. MDE P1 ~\$3/device. Claude ~\$5-10/month.
Cost-free option	\$0	Skip Azure/Sentinel/Defender. Run Splunk-only local stack. Claude API is only paid component.

6. WHAT IS INCLUDED

6.1 Simulation Environment

- **Kali Linux attacker container:** pre-configured with nmap, netcat, Metasploit framework, and custom attack scripts generating realistic malicious traffic patterns.
- **Ubuntu Server victim container:** Running intentionally vulnerable services (SSH, HTTP, FTP) as realistic attack targets, with MDE agent installed for endpoint telemetry.
- **Docker bridge network:** Custom isolated network simulating internal network segmentation between attacker and victim nodes.
- **Attack scenario scripts:** Pre-built scripts for: port scanning, brute-force attempts, lateral movement simulation, and data exfiltration patterns.

6.2 Endpoint Telemetry (Microsoft Defender for Endpoint)

- **MDE Linux agent on victim container:** Captures process creation events, file modification activity, network connection logs, and behavioral alerts from inside the victim OS — providing visibility that Wireshark alone cannot.
- **Defender alert integration:** MDE security alerts flow directly into Sentinel via the native connector, automatically correlating endpoint events with network-level anomalies detected by the transformer model.
- **Device inventory:** The Defender portal maintains a device inventory of all onboarded containers, providing asset context for each incident.

6.3 Detection Pipeline

- **PCAP processing service:** Automated tshark-based capture with rolling file export and Python PCAP parser.
- **Feature engineering module:** Extracts IP addresses, ports, protocols, packet sizes, timing intervals, TCP flags, and session metadata.
- **Transformer model:** Fine-tuned BERT-based model trained on UNSW-NB15 and CICIDS2017 datasets for sequence-based anomaly detection.
- **Model serving API:** FastAPI service exposing /predict endpoint with anomaly score and configurable severity thresholds.
- **Alert engine:** Threshold-based alerting with four severity tiers: Low, Medium, High, and Critical.

6.4 Dual-SIEM Configuration

- **Splunk local SIEM:** Always-on local Splunk Free instance ingesting alert events via HEC, with SPL correlation searches and a real-time threat dashboard. No cost, no time limit.
- **Microsoft Sentinel cloud SIEM:** Azure-hosted Sentinel workspace receiving the same alert stream via Azure Monitor HTTP API, plus Defender for Endpoint telemetry via native connector. KQL analytics rules provide cloud-native detection logic alongside local SPL.
- **Side-by-side comparison:** Both SIEMs receive identical alert data, enabling direct comparison of SPL vs KQL query capabilities, dashboard design differences, and alert correlation approaches.

6.5 SOAR Playbooks (XSOAR)

- **Triage playbook:** Triggered on Medium+ alerts from Splunk webhook. Parses alert metadata, extracts IOCs, and simultaneously queries Sentinel for correlated Defender endpoint events.
- **IOC enrichment playbook:** Queries both VirusTotal (general reputation across 70+ engines) and AbuseIPDB (IP-specific abuse confidence score, number of reports, ISP, country, and usage type — datacenter vs residential). Combines both scores to tag each IOC with a threat category and confidence level.
- **Sentinel correlation playbook:** Pulls related Defender for Endpoint alerts from Sentinel using KQL to enrich the incident with endpoint-level context.
- **Escalation playbook:** Auto-escalates Critical severity cases with MITRE ATT&CK technique tagging based on combined network + endpoint indicators.
- **Claude report generation playbook:** Final step — calls Claude API with the full enriched context (network anomaly score, Defender telemetry, Sentinel events, VirusTotal IOC data, and AbuseIPDB abuse scores) and generates a structured CTI report with BLUF summary, threat assessment, IOC table, and recommended actions.
- **Case closure workflow:** Stores the completed incident in XSOAR and updates the Sentinel incident status via API.

6.6 Claude API Integration

- **Incident report generation:** Claude receives: anomaly score and traffic features, Defender for Endpoint telemetry, Sentinel correlated incidents, VirusTotal IOC enrichment, AbuseIPDB abuse confidence scores, and MITRE ATT&CK tags. Output is a structured BLUF report with threat assessment, IOC table, timeline, and recommended actions.
- **Dual-SIEM narrative:** Claude synthesizes findings from both Splunk and Sentinel into a single coherent incident narrative, highlighting where the two platforms agree or surface different context.
- **Analyst Q&A mode:** Optional interactive mode where the analyst asks Claude follow-up questions about a specific incident using the stored case context.

6.7 Frontend Dashboard

- Real-time alert feed with severity color-coding
- Anomaly score trend chart (time-series)
- Splunk vs Sentinel alert volume comparison panel
- Top attackers and targeted services panel
- MITRE ATT&CK technique heatmap
- Endpoint telemetry summary from Defender
- Incident case list with Claude report viewer
- System health status (all containers, Azure connectivity, API status)

7. IMPLEMENTATION PLAN

Phase 1 - Home Lab Foundation (Weeks 1-2)

Goals

- Install and configure Docker Desktop with WSL2 and resource limits
- Build Kali Linux attacker and Ubuntu Server victim container images
- Create custom Docker bridge network with correct subnet configuration
- Verify attacker-to-victim connectivity and run baseline nmap scan
- Configure Wireshark host capture on Docker bridge interface
- Export first PCAP file and verify packet contents
- Initialize GitHub repository and commit initial scaffold

Deliverables

- docker-compose.yml with attacker and victim services defined
- Dockerfile for custom attacker image with attack tools pre-installed
- First committed PCAP sample
- README: Environment Setup section

Success Criteria

docker-compose up spins up both containers. Wireshark captures traffic on the bridge interface. nmap scan from Kali to Ubuntu completes successfully and appears in PCAP.

Phase 2 - Detection Layer (Weeks 3-4)

Goals

- Download and preprocess UNSW-NB15 and CICIDS2017 datasets
- Build PCAP feature extraction pipeline
- Fine-tune BERT-based transformer model on labeled dataset
- Evaluate model: precision, recall, F1-score, ROC-AUC
- Deploy model as FastAPI inference service in Docker container
- Build alert engine with configurable Low/Medium/High/Critical thresholds

Deliverables

- preprocessing/ module: PCAP parser and feature extractor
- model/ module: training script, weights, evaluation report
- api/ FastAPI service with /predict endpoint and Dockerfile
- README: Detection Layer section

Success Criteria

Feeding Phase 1 PCAP through the pipeline produces anomaly scores. Known attack traffic scores above threshold. Normal traffic scores below threshold.

Phase 3 - Dual SIEM: Splunk + Microsoft Sentinel (Weeks 5-6)

Goals

- Add Splunk Free container to docker-compose.yml and configure HEC
- Connect alert engine output to Splunk HEC
- Write SPL correlation searches for scan, brute-force, and exfiltration patterns
- Build Splunk threat monitoring dashboard
- Create Microsoft Azure free account and activate Sentinel free trial
- Create Azure Log Analytics workspace and enable Microsoft Sentinel
- Configure Azure Monitor HTTP Data Collector API to receive alert forwarding
- Enable Microsoft Defender for Endpoint trial and onboard Ubuntu victim container
- Connect MDE to Sentinel via native Defender data connector
- Write KQL analytics rules in Sentinel mirroring the SPL correlation searches
- Build Sentinel workbook dashboard

Deliverables

- Splunk service in docker-compose.yml with HEC config and SPL searches
- Azure setup guide (Sentinel workspace, Log Analytics, MDE onboarding steps)
- Dual-alert-forwarder script sending events to both Splunk HEC and Azure Monitor API
- KQL analytics rule files committed to repository
- README: Dual SIEM section

Success Criteria

A single simulated attack generates alerts visible in both Splunk dashboard and Microsoft Sentinel. Defender for Endpoint shows the Ubuntu container in device inventory and generates endpoint alerts.

Phase 4 - XSOAR Automation (Weeks 7-8)

Goals

- Add XSOAR Community Edition container to docker-compose.yml
- Configure Splunk webhook to trigger XSOAR on High/Critical alerts
- Install Microsoft Sentinel content pack in XSOAR
- Build triage playbook with IOC extraction and Sentinel query step
- Integrate VirusTotal API for IOC reputation scoring
- Integrate AbuseIPDB API for IP-specific abuse confidence scoring
- Build Sentinel correlation playbook to pull Defender endpoint alerts
- Build escalation playbook with MITRE ATT&CK tagging
- Test end-to-end: Splunk alert → XSOAR → Sentinel query → enriched case

Deliverables

- XSOAR service in docker-compose.yml
- playbooks/ directory with exported playbook YAML files
- Integration configs for Splunk, Sentinel, VirusTotal, and AbuseIPDB
- README: XSOAR SOAR section

Success Criteria

Simulated attack triggers Splunk alert → XSOAR playbook fires → queries Sentinel for Defender telemetry → enriches IOCs via both VirusTotal and AbuseIPDB → creates populated incident case. All automatically.

Phase 5 - Claude Integration & Frontend (Week 9)

Goals

- Add Claude API call as final XSOAR playbook step
- Design report prompt incorporating network anomaly data, Defender telemetry, Sentinel events, VirusTotal enrichment, and AbuseIPDB abuse scores
- Test report quality and BLUF structure across multiple attack scenarios
- Build React frontend dashboard with dual-SIEM alert comparison panel
- Implement Recharts visualizations: alert trends, ATT&CK heatmap, Defender telemetry summary
- Full integration test: packet capture → dual SIEM → XSOAR → Claude report → dashboard

Deliverables

- claude/ module with API integration and prompt templates
- frontend/ React application with all dashboard panels
- Sample Claude-generated incident reports in repository
- README: Claude and Frontend sections

Success Criteria

Full pipeline from simulated attack to Claude-generated CTI report completes within 90 seconds. Report is visible on the React dashboard. Both Splunk and Sentinel dashboards reflect the incident.

Phase 6 - Documentation, Polish & GitHub Portfolio (Week 10)

Goals

- Complete GitHub README with architecture diagram, setup instructions, and demo screenshots
- Record a short demo walkthrough video (optional but highly recommended)
- Write a brief blog post or LinkedIn post about the project
- Ensure docker-compose up spins up the entire local stack cleanly on a fresh machine
- Tag v1.0 release on GitHub

Deliverables

- Final README.md with badges, screenshots, and one-command setup
- CONTRIBUTING.md and LICENSE file
- GitHub release v1.0 with release notes
- LinkedIn post or write-up ready to publish

Success Criteria

Someone unfamiliar with the project can clone the repo, follow the README, and have the local stack running within 30 minutes. The GitHub repository is visually polished and clearly communicates the project's scope.

8. REQUIREMENTS

8.1 Hardware Requirements

Component	Minimum Specification
CPU	Quad-core with Intel VT-x or AMD-V support
RAM	16 GB recommended (12 GB minimum — some services may be limited)
Storage	60 GB free disk space for containers, datasets, models, and logs
OS	Windows 10/11, macOS 12+, or Ubuntu 22.04 LTS
Network	Standard internet connection required for Azure/Sentinel/Defender connectivity

8.2 Software Prerequisites

Software	Notes
Docker Desktop	Enable WSL2 backend on Windows. Allocate 8 GB+ RAM in Docker settings.
Git	Configure with name and email after installation.
Python 3.11+	Check 'Add Python to PATH' during Windows installation.
Node.js LTS	Required for React frontend in Phase 5.
Wireshark	Install on host OS with Npcap (Windows). Not inside Docker.
Claude Code	npm install -g @anthropic-ai/claude-code. Authenticate with Anthropic API key.

8.3 API Keys & Accounts

- **Anthropic API:** console.anthropic.com. Add \$5-10 credit. Used from Phase 5 onward.
- **VirusTotal API:** virustotal.com. Free account. 500 requests/day. Used from Phase 4 onward.
- **AbuseIPDB API:** abuseipdb.com. Free account. 1,000 requests/day. Used from Phase 4 onward for IP reputation scoring.
- **Microsoft Azure:** azure.microsoft.com/free. \$200 credit for 30 days. Required for Phase 3.
- **Microsoft Sentinel:** Activated inside Azure. 31-day free trial. Start at beginning of Phase 3.
- **Microsoft Defender for Endpoint:** Activated inside Azure portal. 30-day trial. Start at beginning of Phase 3.

8.4 Dataset Downloads

- **UNSW-NB15:** research.unsw.edu.au — approximately 2.5 GB. Download before Phase 2.
- **CICIDS2017:** unb.ca/cic/datasets/ids-2017.html — approximately 8 GB. Download before Phase 2.

Note: Store datasets in a datasets/ directory and add them to .gitignore before downloading.

9. CHALLENGES & MITIGATIONS

Challenge	Mitigation Strategy
RAM constraints — running Splunk, XSOAR, and Docker simultaneously	Use Ubuntu Server (no GUI) for victims. Limit Splunk JVM heap to 1 GB. Run model training separately before containerizing inference. Adjust docker-compose memory limits per service.
Azure trial expiration during development	Activate Azure free account and trials only at start of Phase 3. Complete Phase 3 and 4 within the 31-day Sentinel trial window. After trial, reduce Sentinel ingestion volume or pause non-essential data connectors.
MDE agent on Linux container limitations	MDE for Linux is well-supported on Ubuntu 22.04. Follow Microsoft's official onboarding script. Some container-specific behaviors differ from bare-metal — document any gaps found.
Dual-SIEM alert forwarding complexity	Use a lightweight Python forwarder script with two output targets. Implement retry logic and dead-letter queue for failed Azure Monitor API calls.
KQL learning curve alongside SPL	KQL and SPL share conceptual similarities. Write equivalent queries in to reinforce learning. Microsoft Learn has a free KQL module specifically for Sentinel.
Model false positive rate	Tune anomaly threshold per traffic type. Use sliding window approach for sequence scoring. Combine network anomaly score with Defender endpoint signals to reduce false positives.
Claude API cost management	Invoke Claude only on High/Critical severity cases. Cache common report templates. Monitor token usage via Anthropic console. Budget ~\$10/month during active development.
PCAP file size management	Rolling capture with 50 MB max file size. Archive older PCAPs to compressed storage after processing.

10. FUTURE EXPANSION

10.1 Multi-Machine Deployment (Proxmox)

When the HP Pavilion x360 spare laptop receives a RAM upgrade to 16 GB, the simulation layer migrates to Proxmox VMs. Changes required:

- Kali Linux and Ubuntu Server move from Docker containers to Proxmox KVM virtual machines
- pfSense VM added as a virtual firewall between attacker and victim for richer log generation
- MDE agent moves from container to the Ubuntu VM — providing more complete endpoint telemetry
- All service containers (Splunk, XSOAR, transformer API, frontend) remain unchanged on main machine
- Environment variables updated to point at Proxmox VM IPs — single .env file change

10.2 Extended Microsoft Stack

- **Microsoft Purview:** Add data classification and DLP monitoring alongside Defender. Directly addresses the Purview skill gap noted in resume review.
- **Microsoft Defender for Cloud:** Extend Defender coverage to the Azure-hosted components for cloud workload protection.
- **Azure Sentinel SOAR:** Build Sentinel Logic Apps playbooks as a secondary automation layer alongside XSOAR, enabling native Azure automation experience.
- **Microsoft Entra ID:** Add identity and access management telemetry to the lab by simulating authentication events.

10.3 Additional Security Tools

- **ThreatQ:** Add as a threat intelligence platform between XSOAR and the Claude report step, directly mirroring the Leidos ThreatQ implementation project.
- **Suricata IDS:** Add as a network IDS layer on the Docker bridge, feeding alerts into both Splunk and Sentinel alongside transformer model output.
- **Zeek:** Add for richer network metadata: connection logs, DNS, HTTP, and certificate data beyond raw PCAP.
- **MISP:** Integrate as a threat intelligence sharing platform for IOC export and community feed ingestion.
- **Elastic Stack:** Add Elasticsearch and Kibana as a third SIEM option for broadest market coverage.

10.4 Model Improvements

- Multi-class classification (malware, DDoS, port scan, brute-force, exfiltration) rather than binary anomaly detection
- Fusion model combining network anomaly scores with Defender endpoint signals for higher-confidence detections
- Online learning component to adapt model weights as new attack patterns are observed
- SHAP or LIME explainability layer to highlight which features drove each anomaly score

11. REPOSITORY STRUCTURE

The following directory structure will be maintained in the GitHub repository:

```
ai-threat-detector/
├── docker-compose.yml      # Full local lab orchestration
├── .env.example            # Environment variable template
├── .gitignore              # Excludes .env, datasets/, model weights
├── README.md              # Project overview and setup guide
├── docs/
│   └── project-documentation.docx
├── attacker/              # Kali Linux container
│   ├── Dockerfile
│   └── scripts/           # Attack scenario scripts
├── victim/                # Ubuntu Server container
│   ├── Dockerfile
│   └── mde-onboard/       # MDE Linux agent onboarding scripts
├── capture/               # Wireshark / tshark automation
│   └── capture.sh
├── preprocessing/         # PCAP parser and feature extractor
│   ├── pcap_parser.py
│   └── feature_engineer.py
├── model/                 # Transformer model
│   ├── train.py
│   ├── inference.py
│   └── weights/           # Gitignored if large
├── api/                   # FastAPI inference service
│   ├── main.py
│   └── Dockerfile
├── forwarder/             # Dual-SIEM alert forwarder
│   └── forwarder.py       # Sends to both Splunk HEC + Azure Monitor
├── splunk/                # Splunk configuration
│   ├── indexes.conf
│   ├── inputs.conf
│   └── searches/          # SPL query files
├── sentinel/              # Microsoft Sentinel configuration
│   ├── analytics-rules/   # KQL analytics rule definitions
│   ├── workbooks/         # Sentinel dashboard workbook JSON
│   └── setup-guide.md     # Azure/Sentinel/MDE setup walkthrough
├── xsoar/                 # XSOAR playbooks and configs
│   └── playbooks/
├── claude/                # Claude API integration
│   ├── report_generator.py
│   └── prompt_templates/
├── frontend/              # React dashboard
│   ├── src/
│   └── Dockerfile
└── datasets/              # Training data - gitignored
    ├── UNSW-NB15/
    └── CICIDS2017/
```

12. PRE-PROJECT CHECKLIST

Complete every item before writing any code. Items are ordered by dependency.

12.1 Accounts & API Keys

Create all accounts and generate API keys before starting. Store every key in a `.env` file — never hardcode them or commit them to GitHub.

Item	Where / How
Anthropic API key	console.anthropic.com → API Keys → Create key. Add \$5-10 credit.
VirusTotal API key	virustotal.com → Sign up → Profile → API key. Free tier.
AbuseIPDB API key	abuseipdb.com → Sign up → Account → API → Create key. Free tier — 1,000 req/day.
Microsoft Azure account	azure.microsoft.com/free → Create free account. \$200 credit for 30 days. Do NOT activate Sentinel or MDE trials yet — save them for Phase 3.
GitHub repo created	github.com → New repository → ai-threat-detector. Initialize with README and Python .gitignore.
.env file created	Copy .env.example to .env. Paste Anthropic, VirusTotal, and AbuseIPDB keys. Add Azure credentials when ready for Phase 3. Add .env to .gitignore immediately.

12.2 Azure Free Account Setup (Detail)

The Azure free account requires a credit card for identity verification but will not be charged during the free tier period. Follow these steps carefully:

- **Create Azure account:** Go to azure.microsoft.com/free. Click 'Start free.' Sign in with a Microsoft account or create one. Provide credit card for identity verification only — you will not be charged during the free period.
- **Verify \$200 credit:** After signup, go to portal.azure.com → Cost Management + Billing → confirm \$200 credit is showing.
- **Create a Resource Group:** In the Azure portal, create a Resource Group named 'ai-threat-detector-rg' in your nearest region. All project resources will live here for easy cleanup.
- **Do NOT activate trials yet:** Microsoft Sentinel and Defender for Endpoint trials start counting from activation. Do not activate them until you reach Phase 3. Just confirm the Azure account is working.
- **Note your Tenant ID and Subscription ID:** Portal → Azure Active Directory → Overview. Copy Tenant ID and Subscription ID into your `.env` file for later use.

12.3 Software Installation

Install all software on your host machine in the order listed. Restart after Docker Desktop.

Software	Install Steps & Notes
Docker Desktop	docker.com . On Windows: enable WSL2 when prompted. Let initial setup fully complete.
Docker resource config	Docker Desktop → Settings → Resources. RAM: 8 GB+. CPUs: 4+. Disk: 60 GB+. Apply and restart.
Git	git-scm.com . After install: <code>git config --global user.name</code> and <code>user.email</code> .
Python 3.11+	python.org . CRITICAL: check 'Add Python to PATH'. Verify: <code>python --version</code> .
Node.js LTS	nodejs.org . Verify: <code>node --version</code> and <code>npm --version</code> .
Wireshark	wireshark.org . Windows: install Npcap when prompted. Required for Docker bridge capture.
Claude Code	<code>npm install -g @anthropic-ai/claude-code</code> . Verify: <code>claude --version</code> . Authenticate with Anthropic API key.
Azure CLI (optional)	learn.microsoft.com/cli/azure/install-azure-cli . Useful for scripting Azure resource setup in Phase 3.

12.4 Dataset Downloads

Start these downloads before Phase 1 so they are ready for Phase 2.

Dataset	Details
UNSW-NB15	research.unsw.edu.au/projects/unsw-nb15-dataset. ~2.5 GB. Place in datasets/UNSW-NB15/.
CICIDS2017	unb.ca/cic/datasets/ids-2017.html. ~8 GB. Place in datasets/CICIDS2017/.
.gitignore entry	Add 'datasets/' to .gitignore before placing files. Verify with git status.

12.5 Repository Setup

- **Create GitHub repo:** Initialize at github.com with README and Python .gitignore template.
- **Clone locally:** git clone <repo-url>. All work happens inside this directory.
- **Create directory structure:** Create all folders from Section 11. Empty dirs need a .gitkeep file to be tracked.
- **Create .env.example:** Add placeholder entries for all keys. Commit this file — it shows what keys are needed.
- **Create .env:** Copy .env.example to .env and fill in real values. Verify .env is gitignored.
- **Initial commit:** git add . && git commit -m 'Initial scaffold' && git push.

12.6 Docker & Tool Verification

Check	Command & Expected Result
Docker running	docker --version → Docker version 24+
Docker Compose	docker compose version → v2+
Docker functional	docker run hello-world → Hello from Docker!
Python	python --version → Python 3.11+
Node	node --version → v20+
Claude Code	claude --version → version string
Azure CLI (if installed)	az --version → version string

12.7 Pre-Flight Sign-Off

Confirm every item below before moving to Phase 1:

- Anthropic API key created and stored in .env
- VirusTotal API key created and stored in .env
- AbuseIPDB API key created and stored in .env
- Microsoft Azure free account created — \$200 credit confirmed in portal
- Azure Resource Group 'ai-threat-detector-rg' created
- Sentinel and MDE trials NOT yet activated (saving for Phase 3)
- GitHub repository created and initial scaffold committed
- .env added to .gitignore and verified not tracked by Git
- Docker Desktop installed, WSL2 enabled, resource limits increased
- docker run hello-world completes successfully
- Python 3.11+ accessible via python --version
- Node.js LTS accessible via node --version
- Wireshark installed with Npcap — opens without errors
- Git configured with name and email
- Claude Code installed and authenticated
- UNSW-NB15 and CICIDS2017 downloads initiated or complete
- datasets/ directory created and added to .gitignore

13. GLOSSARY

Term	Definition
ATT&CK	MITRE Adversarial Tactics, Techniques, and Common Knowledge — framework for classifying adversary behavior.
AbuseIPDB	A community-driven IP reputation database providing abuse confidence scores, report counts, ISP data, and usage type classification for suspicious IP addresses.
BERT	Bidirectional Encoder Representations from Transformers — pre-trained transformer model adapted for anomaly detection.
BLUF	Bottom Line Up Front — structured writing format where the key conclusion appears first.
CTI	Cyber Threat Intelligence — analyzed information about adversaries, capabilities, and intentions.
CICIDS2017	Canadian Institute for Cybersecurity Intrusion Detection Evaluation Dataset 2017.
Defender for Endpoint	Microsoft's enterprise EDR (Endpoint Detection and Response) platform, formerly Windows Defender ATP.
Docker	Container platform packaging applications and dependencies into isolated, portable units.
EDR	Endpoint Detection and Response — security tooling providing visibility into endpoint activity and automated response.
HEC	HTTP Event Collector — Splunk mechanism for ingesting events via HTTP/HTTPS.
IOC	Indicator of Compromise — an artifact (IP, domain, hash) indicating a system may have been breached.
KQL	Kusto Query Language — the query language used in Microsoft Sentinel and Azure Monitor.
Log Analytics	Azure Log Analytics Workspace — the data store that backs Microsoft Sentinel.
MDE	Microsoft Defender for Endpoint — see Defender for Endpoint.
PCAP	Packet Capture file format used by Wireshark and other network analysis tools.
Sentinel	Microsoft Sentinel — Azure-native cloud SIEM and SOAR platform.
SIEM	Security Information and Event Management — platform for real-time log analysis, alerting, and dashboarding.
SOAR	Security Orchestration, Automation, and Response — platform for automating security workflows.
SPL	Splunk Processing Language — query language used in Splunk for searching and analyzing log data.
UNSW-NB15	University of New South Wales Network Benchmark dataset 2015 — labeled network intrusion dataset.
XSOAR	Extended Security Orchestration, Automation, and Response — Palo Alto Networks' SOAR platform.